

SQL Commands to define tables and constraints defined in the E-R diagram: (Note that attribute names and their data types may be different in your design)

```
CREATE TABLE PROFESSORS(SSN INTEGER, name CHAR(30), age INTEGER,  
    rank CHAR(10), resSpeciality CHAR(50),  
    PRIMARY KEY(SSN))
```

```
CREATE TABLE PROJECTS(ProjNo INTEGER, title CHAR(100), startDate DATE,  
    endDate DATE, budget INTEGER,  
    principalInvestigator INTEGER NOT NULL,  
    PRIMARY KEY(ProjNo),  
    FOREIGN KEY principalInvestigator REFERENCES Professors(SSN))
```

```
CREATE TABLE STUDENTS(StudentID INTEGER, name CHAR(50), age INTEGER,  
    gpa REAL, address CHAR(100),  
    PRIMARY KEY(StudentID))
```

```
CREATE TABLE UG_STUDENTS(StudentID INTEGER, department CHAR(50), year INTEGER,  
    PRIMARY KEY(StudentID),  
    FOREIGN KEY StudentID REFERENCES STUDENTS ON DELETE CASCADE)
```

```
CREATE TABLE GRAD_STUDENTS(StudentID INTEGER, degreeProg CHAR(50),  
    major CHAR(50),  
    PRIMARY KEY(StudentID),  
    FOREIGN KEY StudentID REFERENCES STUDENTS ON DELETE CASCADE)
```

A course is UG_Course or GRAD_Course. Since we have only two choices, no need to create a table for COURSES.

```
CREATE TABLE UG_COURSES(CourseID CHAR(6), CName CHAR(100), credit REAL,  
    PRIMARY KEY(CourseID));
```

```
CREATE TABLE GRAD_COURSES(CourseID CHAR(6), CName CHAR(100), credit REAL,  
    PRIMARY KEY(CourseID));
```

```
CREATE TABLE Enroll(StudentID INTEGER, CourseID CHAR(6), semester CHAR(20),  
    grade CHAR(2),  
    PRIMARY KEY(StudentID, CourseID),  
    FOREIGN KEY StudentID REFERENCES UG_STUDENTS  
    ON DELETE CASCADE,  
    FOREIGN KEY CourseID REFERENCES UG_COURSES)
```

A GRAD_Student can enroll at most 1 UG_Course

```
CREATE TABLE UG_Enroll(StudentID INTEGER, CourseID CHAR(6), semester CHAR(20),
                        grade CHAR(2),
                        PRIMARY KEY(StudentID),
                        FOREIGN KEY StudentID REFERENCES GRAD_STUDENTS
                        ON DELETE CASCADE,
                        FOREIGN KEY CourseID REFERENCES UG_COURSES)
```

```
CREATE TABLE GRAD_Enroll(StudentID INTEGER, CourseID CHAR(6), semester CHAR(20),
                           grade CHAR(2),
                           PRIMARY KEY(StudentID, CourseID),
                           FOREIGN KEY StudentID REFERENCES GRAD_STUDENTS
                           ON DELETE CASCADE,
                           FOREIGN KEY CourseID REFERENCES GRAD_COURSES)
```

```
CREATE TABLE UG_Teach(SSN INTEGER, CourseID CHAR(6), semester CHAR(20),
                       PRIMARY KEY(SSN, semester),
                       FOREIGN KEY SSN REFERENCES PROFESSORS
                       FOREIGN KEY CourseID REFERENCES UG_COURSES)
```

```
CREATE TABLE GRAD_Teach(SSN INTEGER, CourseID CHAR(6), semester CHAR(20),
                          PRIMARY KEY(SSN, semester),
                          FOREIGN KEY SSN REFERENCES PROFESSORS
                          FOREIGN KEY CourseID REFERENCES GRAD_COURSES)
```

```
CREATE TABLE SPONSORS(name CHAR(100),
                       PRIMARY KEY(name))
```

```
CREATE TABLE Support(ProjNo INTEGER, name CHAR(100),
                      PRIMARY KEY(ProjNo, name),
                      FOREIGN KEY ProjNo REFERENCES PROJECTS,
                      FOREIGN KEY name REFERENCES SPONSORS)
```

```
CREATE TABLE WorkOn(ProjNo INTEGER, coInvestigator INTEGER,
                    PRIMARY KEY(ProjNo, coInvestigator),
                    FOREIGN KEY ProjNo REFERENCES PROJECTS,
                    FOREIGN KEY coInvestigator REFERENCES PROFESSORS(SSN))
```

```
CREATE TABLE Work(ProjNo INTEGER, resAssistant INTEGER,
                  PRIMARY KEY(ProjNo, resAssistant),
                  FOREIGN KEY ProjNo REFERENCES PROJECTS,
                  FOREIGN KEY resAssistant REFERENCES GRAD_STUDENTS(StudentID))
```

```
CREATE TABLE Supervise(ProjNo INTEGER, resAssistant INTEGER,  
    supervisor INTEGER,  
    PRIMARY KEY(ProjNo, resAssistant, supervisor),  
    FOREIGN KEY ProjNo REFERENCES PROJECTS,  
    FOREIGN KEY resAssistant REFERENCES GRAD_STUDENTS(StudentID),  
    FOREIGN KEY supervisor REFERENCES PROJECTS(principalInvestigator))
```